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# N9000-13RF

**General Processing Guidelines**

## Cyanate Ester
## Laminate and Prepreg

N8000 is a high-performance advanced material with a high Tg and extremely low Z-Axis expansion, creating a material with signal speed capabilities not achievable through standard epoxy or polyimide.

## Material Handling & Storage

Store laminates flat in a dry environment. Do not bend, scratch or dent laminate.
Store prepreg flat, with a storage temperature of <72° F (<23°C) and ≤50% RH.

## Drilling

N9000-13 RF materials should be drilled using highly polished carbide tools. The use of repointed tools is not recommended. Panels can be drilled in stacks based on total thickness. The total stack height, including entry and backup materials, should not exceed 70% of the effective flute length of the drill bit. For a multilayer PCB, 1 panel per stack is preferred. The use of hard phenolic, rigid entry .020-.032" (0.5 mm - 0.8 mm) and backup .060-.080" (1.5 mm – 2.0 mm) material is recommended. PTFE laminates are much softer than FR4 laminates. The edges of the holes will be distorted if entry and backup materials are not used during drilling. Aluminium or aluminium-clad entry and backup materials are not recommended for N9000-13 RF materials, as aluminium burrs have a tendency to damage the soft PTFE during the machining process. The following drill parameters are recommended as a beginning point to develop specific process parameters.

| Drill bit (mm) | INFEED (IPM) | SPEED (KRPM) | RTR (IPM) | Chip load (mil) | Hit Max |
|---|---|---|---|---|---|
| 0.20 | 73 | 73 | 300 | 1.0 | 500 |
| 0.25 | 89 | 68 | 300 | 1.3 | 500 |
| 0.30 | 85 | 60 | 300 | 1.4 | 500 |
| 0.35 | 83 | 55 | 300 | 1.5 | 500 |
| 0.40 | 87 | 52 | 300 | 1.7 | 500 |
| 0.50 | 96 | 48 | 400 | 2.0 | 500 |
| 0.65 | 74 | 37 | 400 | 2.0 | 600 |
| 0.75 | 64 | 32 | 400 | 2.0 | 600 |
| 1.00 | 48 | 24 | 400 | 2.0 | 600 |
| 1.25 | 40 | 20 | 400 | 2.0 | 600 |

## Hole Cleaning/Deburr

Use proper entry and exit materials to minimize burr formation. If the burrs are small, use a hand pumice scrub followed by high pressure rinsing to remove debris from the holes. Mechanical deburr or scrubbing to remove burrs is not recommended. All loose debris in the holes should be removed prior to plasma treatment.

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## Bake

If the PCB is a hybrid with Nelco N4000-13, then bake after hole cleaning. Bake condition could be 3-4hours @150°C.

## Desmear

PTFE is chemically inert. There is no material available that can chemically remove drill smear from a hole in PTFE. The only way to remove the smear is to use a larger drill bit and cut the smear out of the hole. Plasma processes typically used to desmear epoxy-based boards (O2 / CF4) appear to have limited effectiveness when treating PTFE surfaces. By contrast, NH3 or (70% H2 / 30% N2) blends have been found to yield good results with PTFE as well as sodium etching.

The typical plasma cycle contains three steps: An O2/N2 burn to raise the product temperature, a main process cycle, and an O2 burn to remove ash and residual unreacted resin components. Baking at 30-60min@120°C before Plasma is recommended.

| Step | O2 | N2 | CF4 | H2 | Pressure | RF | Seg Time (min) | Remark |
|---|---|---|---|---|---|---|---|---|
| 1 | 80% | 20% | | | 250Torr | 8000W | 30 | |
| 2 | 80% | 10% | 10% | | 250Torr | 4000W | 10-45 | For FR4 |
| 3 | 100% | | | | 250Torr | 4000W | 10 | |
| 4 | | 30% | | 70% | 250Torr | 4200W | 10-30 | For PTFE |

## Metallization

Skip the chemical desmear and the panels should be processed through metallization within 8 hours after plasma. If the hold time is exceeded, re-activate the PTFE using step 1 and step 4 from the list above.

## Imaging

The dimensional stability of Neltec's N9000-13 RF material is excellent. The dimensional change should be <0.5 mils/inch (0.005 mm/cm), depending on the construction and process.

## Solder Mask

If the PTFE is in the outer layer, the solder mask should be applied within 24 hours after etching; the tooth of the copper provides mechanical bonding of the mask to the PTFE surface. If the hold time is exceeded, re-activate the PTFE using step 1 and step 4 from the list above.

## Reflow/HASL

Bake prior to re-flow or hot air solder leveling.

## Developing/Etching/Stripping

Due the low modulus of the PTFE materials, frames and leaders may be required when processing thin laminates through some conveyorized equipment. Standard etchants can be used.

## Routing

Routing, steel rule die cutting, waterjet cutting, and laser cutting can be used depending on required edge quality and cutting tolerances. In the case of routing, standard carbide (helical two flutes or four flutes) routers are recommended. Copper foil should be etched away from the routed edge to prevent burring. For edge plating, follow the hole plating recommendations above.

# Appendix

*If the out layer of the PCB is FR-4, not PTFE, then the entry and back-up materials for drilling can be aluminum or aluminum-clad. In this case, mechanical deburring can be applied.

*If the PCB is a multilayer hybrid with the Nelco N4000-13 series, the peak temp. for lamination should be less than 195°C, and the cure time should be less than 110 min.

These guidelines can provide only basic and reference information for PCB fabricators. Because of different environment, equipment, tooling and so on, in all instances, the user shall determine suitability in any given conditions or applications. For more detailed processing information, please contact with the AGC engineer or sales representative.